

DIVIT RAWAL

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Research Interests. Statistical learning theory and high-dimensional statistics, learning under geometric structure, training dynamics and optimization in deep networks.

EDUCATION

University of California, Berkeley

2023 – 2027

B.A. in Statistics, Computer Science, and Physics

Berkeley, CA

Relevant Coursework: Statistical Learning Theory[†]; Theoretical Statistics[†]; Mathematical Statistics[†]; Optimization; Learning for Dynamics and Control[†]; Probability and Random Processes; Real Analysis; Randomized Linear Algebra, Optimization, and Large-Scale Learning[†]. ([†] denotes graduate-level coursework)

Teaching: EE 122 (Communication Networks) Head TA (Sp26), Reader (Sp25). Developed lab curriculum, wrote and graded exams and homeworks, and led weekly recitation sections.

PUBLICATIONS AND PREPRINTS

($\alpha\beta$) denotes alphabetical ordering; * denotes equal contribution.

[1] **Majority-of-Three is Optimal**

Divit Rawal, Nikita Zhivotovskiy. ($\alpha\beta$)

Under review, TMLR. [\[arXiv\]](#)

[2] **A Theory of Saddle Escape in Deep Nonlinear Networks**

Divit Rawal, Michael R. DeWeese.

Under review, NeurIPS 2026. [\[arXiv\]](#)

[3] **Rao-Blackwellized Score Matching on Manifolds**

Divit Rawal.

ICML 2026, Workshop on Structured Probabilistic Inference and Generative Modeling. [\[arXiv\]](#) / [\[poster\]](#)

[4] **Minimax Rates for Hyperbolic Hierarchical Learning**

Divit Rawal, Sriram Vishwanath.

Preprint. [\[arXiv\]](#)

[5] **ALPHANSO: Open-Source Modeling of (α, n) Neutron Source Terms**

Divit Rawal, Anthony J. Nelson, William Zywiec, Daniel Siefman.

Nuclear Instruments and Methods in Physics Research, Section A. [\[doi\]](#)

Also presented at ANS Student Conference 2026 and INMM Annual Meeting 2026.

AWARDS AND GRANTS

Best Paper Award, American Nuclear Society Student Conference

Apr 2026

Best Paper in Mathematics, Computation, and AI Applications for “ALPHANSO: Open-Source Modeling of (α, n) Source Terms,” selected from ~ 250 submissions.

VESSL AI Academia Grant

Aug 2025

Awarded for independent research into embeddings and information geometry; sole undergraduate recipient among cohort of Ph.D. students and postdoctoral researchers. (Supported work on [3] and [4].)

RESEARCH EXPERIENCE

Berkeley Statistical Machine Learning Group

Apr 2026 – Present

Researcher (advisor: Nikita Zhivotovskiy)

Berkeley, CA

Resolved open conjecture on the optimality of majority of three ERMs via conditional probability tail bounds; authored resulting paper [1].

Extending previous results on majority voting from the realizable PAC setting to the agnostic PAC setting to achieve optimal sample-complexity rates.

Berkeley Artificial Intelligence Research (BAIR)

Sep 2024 – Present

Researcher (advisor: Michael DeWeese)*Berkeley, CA*

Authored paper [2] deriving an exact layer-imbalance identity for deep nonlinear networks and a critical-depth escape law, with matching theory and simulation under both symmetric and He-normal initialization.

Analyzed in-context learning in kernel ridge regression via spectral analysis of kernel matrices (eigenlearning), deriving mode-wise error dynamics and linking adaptation to spectral bias and target alignment.

Lawrence Livermore National Laboratory

Mar 2025 – Present

Researcher, Nuclear Science and Security Consortium Fellow (advisor: Daniel Siefman)*Livermore, CA*

Developed ALPHANSO [5], an open-source framework for deterministic modeling of (α, n) neutron production using continuous slowing-down approximations and modern evaluated nuclear data.

Benchmarked stochastic and deterministic transport predictions against legacy codes and experimental datasets, demonstrating improved accuracy and extensibility relative to legacy pipelines.

INVITED TALKS

ALPHANSO: Open-Source (α, n) Neutron Source Terms

Lawrence Livermore National Laboratory

Mar 2026

Lawrence Berkeley National Laboratory

Mar 2026

University of Tennessee, Knoxville

Jun 2026

INDUSTRY EXPERIENCE

Cisco Systems, Foundation AI

Jun – Aug 2025

*Software Engineer Intern**San Francisco, CA*

Contributed to post-training and evaluation of cybersecurity-specialized LLMs built on Llama 3.1 8B, including instruction tuning and preference alignment (RLHF).

Supported open-weight releases Foundation-Sec-8B and Foundation-Sec-8B-Instruct; contributed to accompanying technical reports on data curation, training methodology, and benchmark evaluation.

ExperienceFlow AI

May – Sep 2024

*Machine Learning Engineer Intern**Remote*

Modeled FSM transition dynamics using transformer, state-space, and graph architectures; evaluated generalization and sample efficiency in low-label sequence prediction settings.

Developed RL-based control policies (DQN, SARSA) for dynamic state reasoning systems, improving stability and convergence.

Amazon

Aug – Dec 2023

*Software Engineer Intern**Remote*

Diagnosed and resolved production pipeline failures affecting 1M+ users; implemented a k -means clustering module in Java for OpenSearch ml-commons.

TECHNICAL SKILLS

Programming: Proficient in Python, C++, Java, SQL; intermediate in MATLAB, FORTRAN, Julia, R.

Scientific Computing: NumPy, Numba, SciPy, Monte Carlo methods, randomized algorithms.

ML/AI: PyTorch, JAX, TensorFlow.